

Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Progress update

Andrés Hojman Marigen Narea Jorge Rodríguez Pedro Carneiro Micaella Magnasco

Pontificia Universidad Católica de Chile

October 2026

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

Design innovation

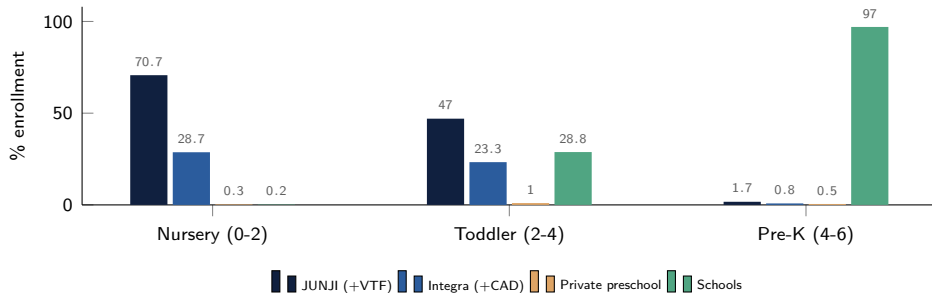
First sub-municipal panel linking public preschools (JUNJI/Integra), the Census/RSH, and geolocation at the neighborhood-unit (UV) level — no prior study in the region has this level of spatial detail.

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Identification: staggered-adoption event study

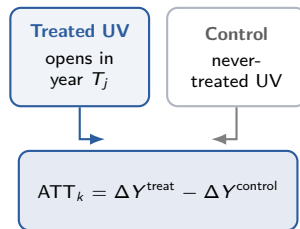
Design: staggered-adoption difference-in-differences with event study

Unit of treatment: neighborhood unit (UV) — finer geographic resolution than comparable literature

Unit of analysis: panel of women, linked to their UV of residence

Fixed effects: UV and year (baseline); since the panel tracks the same women over time, **woman fixed effects** can replace UV effects as a robustness check

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026) — exposure to greater *potential* access, not necessarily individual take-up



Compares the *change* for treated UVs against the change
for never-treated UVs — not levels.

A UV is treated when it opens its first preschool

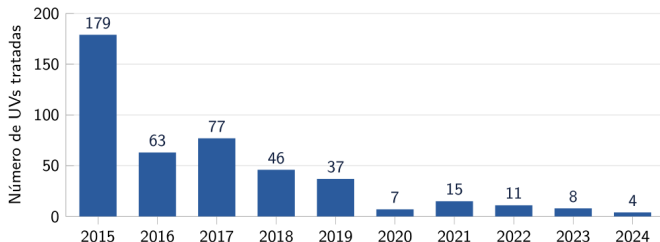
Treatment definition

A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,265	33.0%
Never-treated	0 in 2014, stays at 0 through 2024	4,145	60.5%
Treated	0 in 2014, opens first center 2015–2024	447	6.5%
Total universe		6,857	100%

Source: base_cobertura_cp.dta. Design sample (muestra_t1==1) = Never-treated + Treated = 4,592 UVs.

Treatment cohorts by year



447 UVs tratadas (muestra: 0 centros en 2014). El 71 % (319) se concentra entre 2015 y 2017.

Fuente: elaboración propia sobre base_cobertura_cp.dta.

Using **2014 as the reference year** captures the full 2015 cohort (179 UVs, **40%** of all treated) — the largest cohort, tied to the **Bachelet administration's** preschool expansion policy. Without 2014 as the base year, that cohort cannot be identified as “went from 0 to 1” (no prior year to compare), and treated UVs would drop to **~268** instead of 447.

Empirical specification and estimator

$$Y_{iut} = \lambda_u + \lambda_t + f(\text{Age}_{it}) + \sum_{k \neq -1} \beta_k \mathbb{1}\{t - G_u = k\} + \varepsilon_{iut}$$

Y_{iut} : outcome for woman i , UV u , year t — **one observation per woman-year** (not collapsed to UV-year).

λ_u, λ_t : UV and year fixed effects (primary specification).

$f(\text{Age}_{it})$: flexible age control (labor market spec also controls for the youngest eligible child's age).

β_k : ATT k years before/after opening, relative to $t = -1$ (omitted reference).

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (*JEL*, 2026).

Parallel trends tested via pre-period coefficients ($k < 0$); main threat is non-random timing of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile) — tested directly ahead. **Woman FE** (μ_i replacing λ_u) run only as a **robustness check** — absorbs individual-level composition, but does not address *where/when* preschools are built, which remains a UV-level question.

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV × year) Women panel (base)	JUNJI/Integra transparency requests + UV shapefile (MDSF) RSH (2016–2024) + FPS (2014–2015)	n_total, tratada, g1, grupo_t1, rel_t1 id_uv_2024, pob_total_uv, densidad_ninos_uv, cambia_uv
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	ingreso_anual, meses_trabajando, trabaja, dep./indep.
Fertility	+ Civil Registry birth records (via RIS)	tuvo_hijo, n_hijos_menor5
First stage (enrollment)	+ Matrícula parvularia (child-level)	matriculado, tasa_matricula

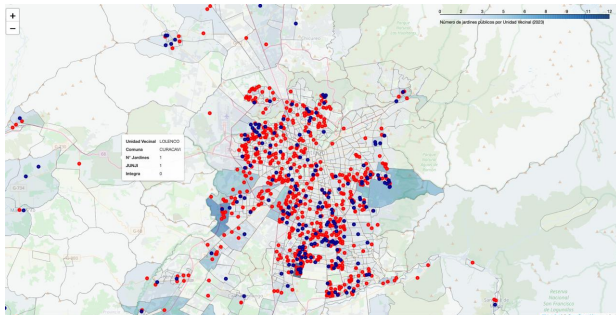
All sources are **administrative registries** (JUNJI/Integra, SII, AFP, RSH/FPS) — no survey self-report used for the core outcomes.

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is spatially matched to the UV where it is located (GeoPandas spatial join)

Uses the **2024 UV cartography** as a unified boundary reference across all years

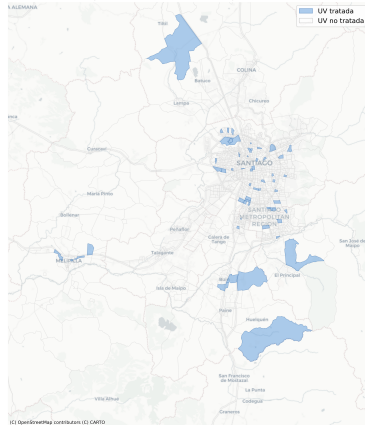
This determines the UV each person corresponds to in 2024 — the basis for treatment assignment



Where the treated UVs are located

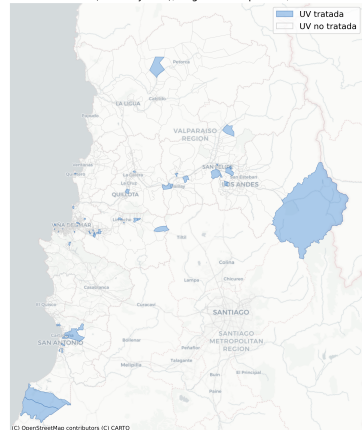
Treated UVs, RM

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Gran Santiago
(Tiltil - San Francisco de Mostazal), 2015-2024



Treated UVs, Valparaíso

UVs tratadas ($0 \rightarrow \geq 1$ jardín), Región de Valparaíso, 2015-2024



Treated UVs, 2015–2024 (go from 0 to 1+ preschools). Source: base_cobertura_cp.dta.

Who is in each sample, and what we measure

Samples

Fertility: women of fertile age (18–49)

Labor market: women with a child < 5 *that specific year* — not “ever had a child”

First stage: children of treatment-sample mothers, birth to birth+4

Outcome variables

Fertility: had a child that year (`tuvo_hijo`), from Civil Registry birth records

Labor market: annual income (`renta` + `cotizaciones` fallback), months worked, employed (yes/no), dependent vs. independent

Income variable

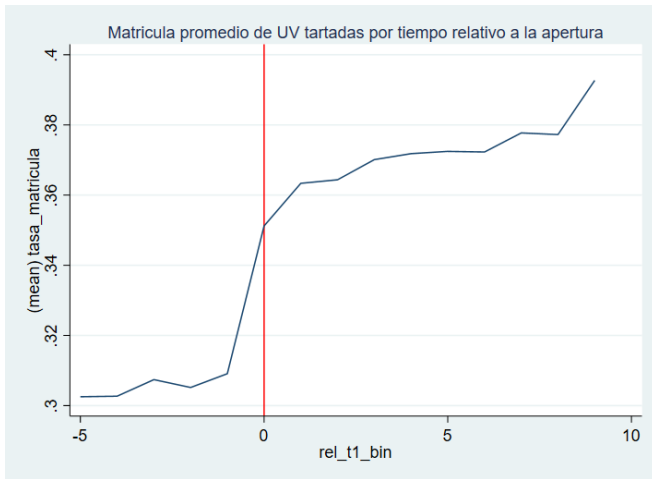
Two **administrative** sources only: `renta imponible (SII)` + `cotizaciones (AFP)` as fallback. Decision made **not** to add self-reported RSH income — subreporting risk in a means-tested instrument.

Treated and never-treated UVs look alike — except in size

Variable	Unweighted		Norm. diff.	Sig. ^a
	Never-tr.	Treated		
Annual income (M CLP)	3.60	3.55	-0.033	*
Months worked	8.14	8.10	-0.024	*
Age	29.24	29.25	0.008	
N. children <5	1.11	1.11	0.045	
Hours worked	37.07	37.68	0.099	
Works (share)	0.45	0.45	0.023	
Education (code)	4.98	5.02	0.028	
UV population	1126.79	2011.08	0.410	***
Child density 0-5 UV	0.06	0.06	0.155	***
N (UVs)	3,909 / 422			

^a t-test (* $p < 0.05$, *** $p < 0.01$), normalized diff. prioritized. Year 2014 (pre-treatment). Only **UV population** (0.41) and **child density** (0.16) exceed the 0.25 concern threshold — selection is in *where* preschools are placed, not *who* lives there → motivates UV + year FE. Source: `tabla_balance_uv_2014.pdf`.

Enrollment jumps sharply the year a preschool opens



The enrollment effect is robust across three estimators

k	Unweighted TWFE	Weighted TWFE	Sun-Abraham
-5	-0.002	-0.015	0.006
-4	-0.009	-0.011*	0.004
-3	-0.013 [†]	-0.018***	-0.005
-2	-0.023***	-0.017***	-0.017***
0	0.020***	0.007*	0.023***
1	0.023***	0.013***	0.026***
2	0.016***	0.012***	0.019***
3	0.018***	0.007**	0.021***
4	0.017***	0.009**	0.020***
5	0.019***	0.009***	0.023***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, [†] $p = 0.066$. Base rate $\approx 32.6\%$. $N = 48,348$ UV-year, clusters = 4,568 UVs.

Effect: $\approx 2\text{pp}$ on a $\approx 33\%$ base ($\approx 6\%$ relative), robust across specifications and growing slightly over time. Source: `primera_etapa.txt`.

The honest problem: $k = -2$ does not vanish

$k = -2$ is **negative and significant in all three specifications**, and **survives Sun–Abraham** — the estimator designed to remove cross-cohort comparison bias

By contrast, $k = -3$ loses significance under Sun–Abraham ($p=0.066 \rightarrow p=0.429$)

Hypotheses under investigation (none confirmed)

1. **Registration lag / construction transition**: the recorded opening year may lag actual full-capacity operation
2. **Age-cutoff measurement** (edad_30_06): possible source of noise, no confirmed mechanical link to $k = -2$ specifically
3. **Partial COVID overlap**: weak, covers only cohorts with $g1 \in [2021, 2023]$

∅ **Placeholder**: Disaggregation of $k = -2$ by opening cohort ($g1$) — running before Thursday.

Labor market outcomes



PLACEHOLDER — results pending

Income and employment estimates (corrected
UV + year FE specification) — in progress

Sample: women with a child <5 that year. Outcomes: annual income, months worked. Two administrative income sources (renta + cotizaciones), $\approx 47\%$ missing, non-differential by treatment.

Fertility outcomes



PLACEHOLDER — results pending

Fertility (*tuvo_hijo*) event-study estimates — in progress

Sample: women 18–49. Outcome: whether the woman had a child that year, at the $UV \times \text{year}$ level.

Next steps

Immediate

Finish labor market estimation
(correct year-FE spec)

Finish fertility estimation

Short term

Disaggregate $k = -2$ by opening cohort

Heterogeneity by education level
(nursery/toddler vs. pre-K)

Medium term

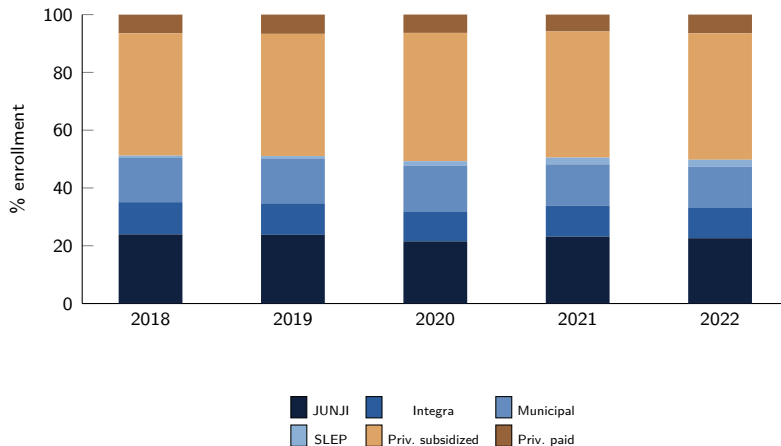
Robustness: weighted vs.
unweighted, outlier UVs

Placebo: pre-treatment birth rate

Write results section

Appendix

Appendix A1. Enrollment by dependency, 2018–2022



Source: Tabla 14, Matrícula Oficial Educación Parvularia por dependencia 2018–2022, Centro de Estudios MINEDUC.

Appendix A2. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat:** JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A3. Weighted vs. unweighted robustness

Weighted specification (by `n_ninos`) shows a visibly worse pre-trend: 3 of 4 pre-period coefficients significant, vs. 1 of 4 unweighted

Consistent with a small number of large outlier UVs dominating the weighted average:
`n_ninos` median 40, mean 78, p99 541, max $\approx 3,200$

Pending: robustness check excluding or capping the largest UVs

Appendix A4. Data limitation: UV assignment in 2014–2015

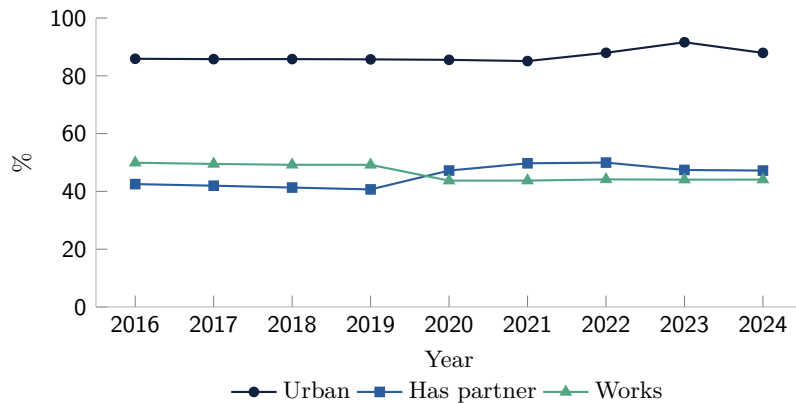
FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A5. Panel composition over time



Panel of women aged 18–70 (RSH), 2016–2024.

Appendix A6. Task status

Block	Task	Status	Note
2. Data generation	Coverage base (preschools $\times$ UV, 2014–2024)	ready	Verified; private preschools (transparency) still a possible extension
	Women panel (2014–2024, incl. FPS anchor)	ready	2014–2015 anchored to 2016 UV (see A5)
	Rentas / Cotizaciones cleaning	advanced	Independent/state/FF.AA. cotizaciones still pending
	Children registry cleaning	ready	tuvo_hijo, n_hijos_menor5
	Merge panel + outcome variables	ready	By rut_inn
	Treated / non-treated definition	ready	grupo_t1, g1, muestra_t1
3. Panel analysis	First-stage estimation (enrollment)	ready	3 specifications; $k = -2$ open issue
	Balance / validity check	ready	UV-level balance table
4. Data analysis	Methodology choice	ready	Sun–Abraham within Baker et al. (JEL 2026) framework
	Treatment definition	ready	Extensive margin, $0 \rightarrow 1$
	Run regressions (labor, fertility)	in progress	Expected before Thursday
	Analyze results	in progress	First stage analyzed; rest pending
5. Mechanisms	Heterogeneity by group / level	not started	
	Public-policy implications	not started	
6. Writing	—	in progress	